



BAULKHAM HILLS HIGH SCHOOL

JUNE 2012

YEAR 12 ASSESSMENT TASK 3

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in questions 1- 4
- Marks may be deducted for careless or badly arranged work
- Standard integrals are provided on page 7

Total marks – ~~46~~ 44

Exam consists of 7 pages.

Question 1

Marks

a) Find the indefinite integrals:

i) $\int \frac{\tan^{-1} x}{1+x^2} dx$ 2

ii) $\int \frac{dx}{\sqrt{x^2+2x+5}}$ 2

iii) $\int x\sqrt{x+1} dx$ 3

b) i) If $\frac{7x^2-3x+2}{(x-2)(x^2+x+2)} = \frac{A}{x-2} + \frac{Bx+C}{x^2+x+2}$ 2

Find A, B and C

ii) Hence determine 2

$$\int \frac{7x^2-3x+2}{(x-2)(x^2+x+2)} dx$$

c) Evaluate

$\int_0^{\frac{\pi}{2}} \frac{4 d\theta}{3+5\cos\theta}$ 3

a) Let n be a positive integer and let

$$I_n = \int_1^3 (\ln x)^n dx$$

i) Prove that $I_n = 3(\ln 3)^n - nI_{n-1}$

2

ii) Hence determine

$$\int_1^3 (\ln x)^2 dx$$

3

b) The base of a solid is a parabolic segment of $y = x^2$ cut off by the chord $y = 4$

5

Each plane sector of the solid perpendicular to the axis of the parabola is a

rectangle whose base is a chord and its height is $\frac{1}{2}(4 - y)$. Find its volume.

Question 3 - Start a new page

Marks

- a) i) Prove that

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

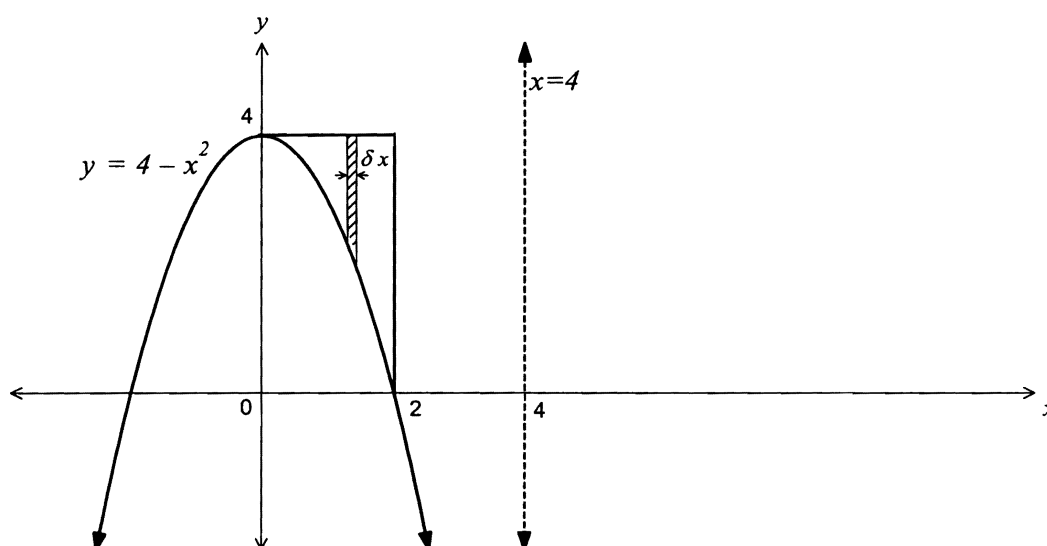
2

- ii) Use this to evaluate

$$\int_0^\pi \frac{x \sin x}{1 + \cos^2 x} dx$$

4

- b) The area bounded by $y = 4 - x^2$, $x = 2$ and $y = 4$ is rotated about the line $x = 4$. Using the method of cylindrical solids



- i) Show that the volume of a cylindrical shell of thickness δx is

$$\pi x^2 (8 - 2x) \delta x$$

2

- ii) Find the volume of the solid.

3

a) i) Show that

$$\int_1^2 \frac{1}{x^2} \ln(x+1) dx = \frac{1}{2} \ln \frac{4}{3} + \int_1^2 \frac{1}{x(x+1)} dx$$

2

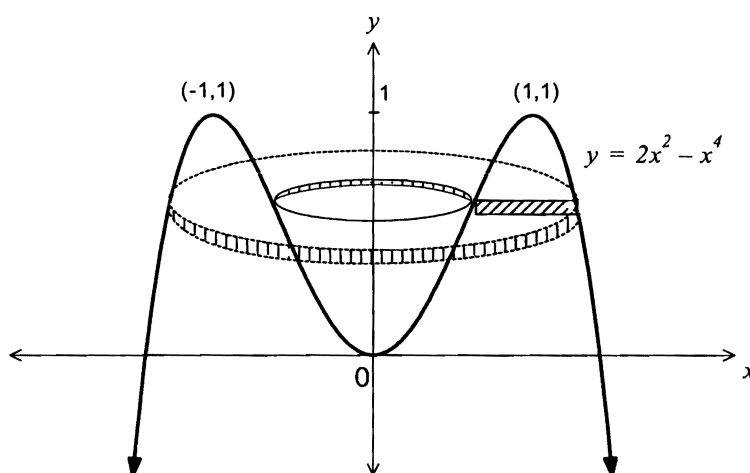
ii) Hence evaluate

$$\int_1^2 \frac{1}{x^2} \ln(x+1) dx \text{ in its simplest form}$$

3

b) The region bounded by the curve $y = 2x^2 - x^4$ and $y = 0$ is rotated about the y -axis, as shown in the diagram below.

4

i) Show that the cross sectional area of a slice perpendicular to the y -axis is

2

$$A = 2\pi\sqrt{1-y}$$

ii) Find the volume formed.

2

~ End of Exam ~

a) 1) $\int \frac{t^{n-1} x}{1+x^2} dx$

Let $u = t^{n-1} x$

$\therefore du = \frac{1}{1+x^2} dx$

$I = \int u du$

$= \frac{u^2}{2} + C$

$= \frac{1}{2} (t^{n-1})^2 + C$

(2)

ii) $\int \frac{dx}{\sqrt{x^2+2x+5}}$

$= \int \frac{du}{\sqrt{(u+1)^2+4}}$

$= \ln(x+1 + \sqrt{x^2+2x+5}) + C$

(2)

iii) $\int x \sqrt{x+1} dx$

Let $u = x+1$

$\therefore du = dx$

$\therefore I = \int (u-1) u^{\frac{1}{2}} du$

$= \int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du$

$= \frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} + C$

$= \frac{2}{5} (x+1)^{\frac{5}{2}} - \frac{2}{3} (x+1)^{\frac{3}{2}} + C$

$u = \frac{2}{5} (x+1) \sqrt{x+1} - \frac{2}{3} (x+1) \sqrt{x+1} + C$

(3)

b) 1) $7x^2 - 3x + 2 = A(x^2 + x + 2) + (Bx + C)(x-2)$

$x=2 \Rightarrow 8A = 24$

$\therefore A = 3$

coef of $x^2 \Rightarrow A + B = 7$

$B = 4$

and the $2A - 2C = 2$

$C = 2$

(2)

ii) $\therefore I = \int \frac{3}{x-2} + \frac{4x+2}{x^2+x+2} dx$

$= 3 \ln(x-2) + \int \frac{2(2x+1)}{x^2+x+2} dx$

$= 2 \ln(x-2) + 2 \ln(x^2+x+2) + C$

(2)

c) $\int_0^{\frac{\pi}{3}} \frac{4}{3+t \tan \theta} dt$ Let $t = \tan \theta$

$= \int_0^1 \frac{4}{3+t^2} \cdot \frac{1}{1+t^2} dt$ $du = \frac{2dt}{1+t^2}$

$= \int_0^1 \frac{8}{8-t^2} dt$ $a=0$ $b=1$

$= \int_0^1 \frac{4}{4-t^2} dt$

$= \int_0^1 \frac{1}{2-t} + \frac{1}{2+t} dt$

$= \left[-\ln(2-t) + \ln(2+t) \right]_0^1$

$= \ln 3$

(3)

$$2.) \quad 1) \quad I_n = \int_1^3 (ln x)^n dx$$

$$\text{let } u = (ln x)^n \quad v' = 1$$

$$u' = n(ln x)^{n-1} \cdot \frac{1}{x} \quad v = x \quad \perp$$

$$I_n = \left[x(ln x)^n \right]_1^3 - n \int_1^3 (ln x)^{n-1} \cdot x \cdot \frac{1}{x} dx$$

$$I_n = 3(ln 3)^n - n I_{n-1} \quad \perp \quad (2)$$

$$1) \quad I_2 = \int_1^3 (ln x)^2 dx$$

$$I_2 = 3(ln 3)^2 - 2 I_1$$

$$I_1 = 3 ln 3 - I_0 \quad \perp$$

$$I_0 = \int_1^3 dx$$

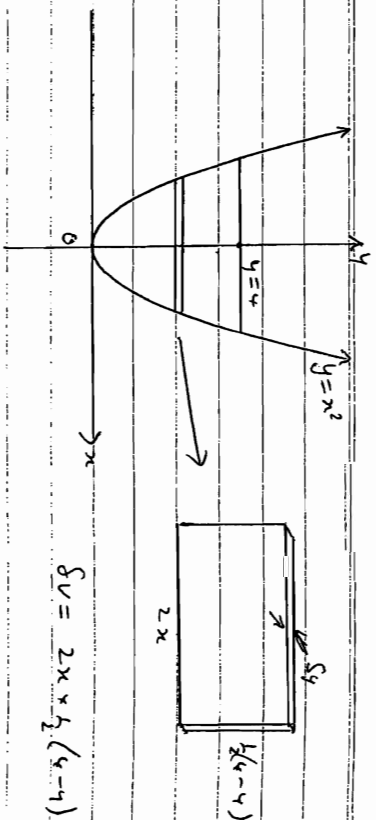
$$= 2$$

$$\therefore I_1 = 3 ln 3 - 2 \quad \perp$$

$$I_2 = 3(ln 3)^2 - 2(3 ln 3 - 2) \quad (3)$$

$$= 2(ln 3)^2 - 6 ln 3 + 4 \quad \perp$$

2b)



$$\delta v = 2x \cdot \frac{1}{2} (4 - y) \cdot \delta y$$

$$= x(4 - y) \delta y \quad \perp$$

$$v = \sum_{y=0}^4 x(4 - y) \delta y$$

$$\perp$$

$$= \int_0^4 x(4 - y) dy$$

$$= \int_0^4 y^{\frac{1}{2}} (4 - y) dy \quad \perp$$

$$= \int_0^4 \left(4y^{\frac{1}{2}} - y^{\frac{3}{2}} \right) dy$$

(5)

$$= \left[\frac{8}{3} y^{\frac{3}{2}} - \frac{2}{5} y^{\frac{5}{2}} \right]_0^4 \quad \perp$$

$$= \frac{64}{3} - \frac{64}{5}$$

$$= \frac{128}{15} \text{ units} \quad \perp$$

$$4a) \quad I = \int_1^3 \frac{1}{x^2} \ln(x+1) dx \quad \text{Let } u = \ln(x+1) \quad u' = x^{-2}$$

$$u' = \frac{1}{x^2} \quad v = -x^{-1}$$

$$\therefore I = - \left[\frac{\ln(x+1)}{x} \right]_1^3 + \int_1^3 \frac{dx}{x(x+1)}$$

$$= - \left[\frac{\ln 3}{2} - \frac{\ln 2}{2} \right] + \int_1^3 \frac{dx}{x(x+1)}$$

$$= \frac{\ln 2 - \ln 3}{2} + \int_1^3 \frac{dx}{x(x+1)}$$

$$= \frac{1}{2} \ln 2 - \frac{1}{2} \ln 3 + \int_1^3 \frac{dx}{x(x+1)}$$

$$= \frac{1}{2} \ln 2 - \frac{1}{2} \ln 3 + \int_1^3 \frac{dx}{x(x+1)}$$

$$\text{Let } I_2 = \int_1^3 \frac{dx}{x(x+1)}$$

$$= \int_1^3 \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$= \left[\ln x - \ln(x+1) \right]_1^3$$

$$= (\ln 2 - \ln 3) - (\ln 1 - \ln 2)$$

$$= 2\ln 2 - \ln 3$$

$$\therefore I = - \frac{1}{2} \ln 2 + 2\ln 2 - \ln 3$$

$$= 3\ln 2 - \ln 3$$

(3)

b)

$$A = \pi(R^2 - r^2)$$

radius R, r are the radii of circles

$$y = 2x^2 - x^4 \quad \text{radius } y \text{ is cent.}$$

$$x^4 - 2x^2 + y = 0$$

$$x^2 = \frac{2 \pm \sqrt{4-4y}}{2}$$

$$= 1 \pm \sqrt{1-y}$$

$$R^2 = 1 + \sqrt{1-y} \quad r^2 = 1 - \sqrt{1-y}$$

$$\therefore R^2 - r^2 = 1 + \sqrt{1-y} - 1 + \sqrt{1-y}$$

$$= 2\sqrt{1-y}$$

$$\therefore A = 2\pi\sqrt{1-y}$$

$$\delta v = 2\pi\sqrt{1-y} \delta y$$

$$\therefore v = 2\pi \int_0^{\sqrt{1-y}} \delta y$$

$$= \left[2\pi \cdot \frac{-2}{3} (1-y)^{\frac{3}{2}} \right]_0^{\sqrt{1-y}}$$

$$= -\frac{4\pi}{3} [0 - 1]$$

$$= \frac{4\pi}{3} \text{ cc units.}$$

(2)

$$3) \text{ Given } \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\text{Let } u = a-x$$

$$x=a$$

$$u=0$$

$$x=0$$

$$u=a$$

$$du = -dx$$

$$\int_0^a f(x) dx = \int_a^0 -f(a-u) du$$

$$= \int_0^a f(a-u) du$$

(2)

$$1) I = \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$$

$$x(\pi-x) = \pi-x$$

$$\cos^2(\pi-x) = (-\cos x)^2 = \cos^2 x$$

$$= \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x)}{1+\cos^2(\pi-x)} dx$$

$$\therefore I = \int_0^{\pi} \frac{(\pi-x) \sin x}{1+\cos^2 x} dx$$

$$I = \pi \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$$

$$2 \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = \pi \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$$

$$\text{Let } u = \cos x$$

$$x=0, \cos x = 1$$

$$du = -\sin x dx$$

$$2I = \pi \int_1^{-1} \frac{-du}{1+u^2}$$

(4)

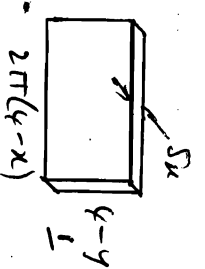
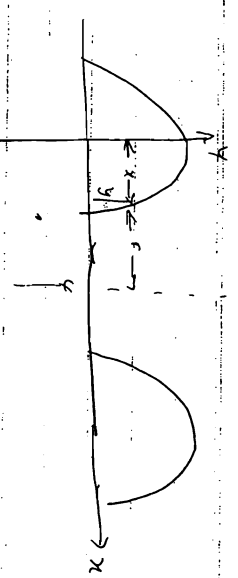
$$= \pi \int_{-1}^1 \frac{du}{1+u^2}$$

$$= \pi \left[\tan^{-1} u \right]_{-1}^1$$

$$= \pi \left[\tan^{-1} 1 - \tan^{-1} (-1) \right] = \pi^2$$

$$\therefore I = \frac{\pi^2}{4}$$

b



$$\delta V = 2\pi r h$$

$$\text{where } h = y, r = 4-x$$

$$\therefore \delta V = 2\pi (4-x) y \delta x$$

$$y = x^2$$

$$\therefore \delta V = 2\pi x^2 (4-x) \delta x$$

$$= \pi x^2 (8-2x) \delta x$$

(5)

$$1) V = \pi \int_0^2 (8x^2 - 2x^3) dx$$

$$= \pi \left[\frac{8x^3}{3} - \frac{2x^4}{4} \right]_0^2$$

$$= \pi \left[\frac{64}{3} - 8 \right]$$

$$= \frac{40\pi}{3} \text{ units}^3$$

(3)